

Question 1

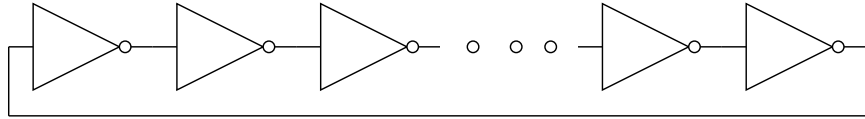
Estimate t_{pd} for a unit inverter driving m identical unit inverters. If the driver is w times unit size, what is the delay? If a unit transistor has $R=10\text{ k}\Omega$ and $C=0.1\text{ fF}$ in a 65 nm process, compute the delay, in picoseconds, of the inverter with a fanout of $h=4$.

Question 2

Use the linear delay model to estimate the delay of the fanout-of-4 (FO4) inverter. Assume the inverter is constructed in a 65 nm process with $\tau=3$ ps.

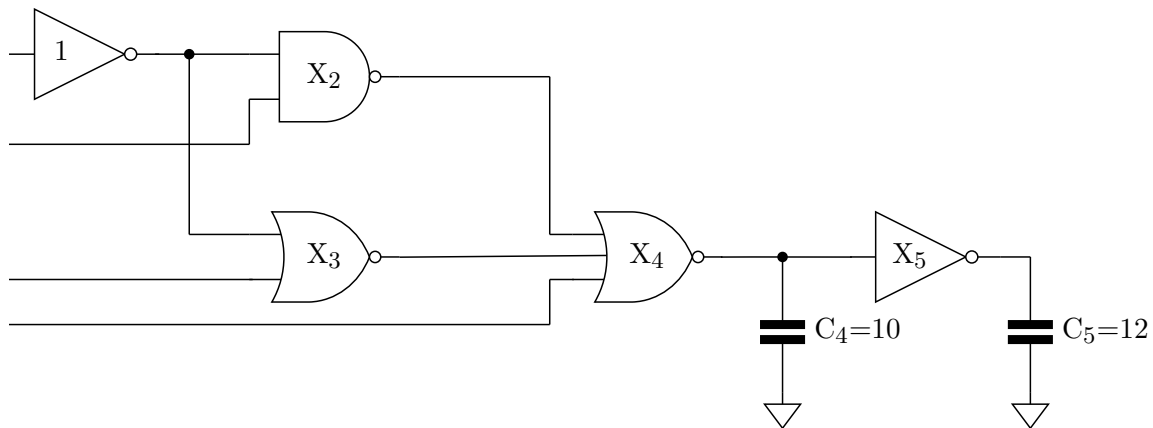
Question 3

A ring oscillator is constructed from an odd number of inverters, as shown in Figure. Estimate the frequency of an N-stage ring oscillator. What will be the frequency of 31 stage ring oscillator if $\tau=3$ ps in a 65 nm process?



Question 4

The circuit in Figure has nonuniform branching, reconvergent fanout, and a wire load in the middle of the path, all of which stymie back-of-the-envelope application of Logical Effort. The wire load is given in the same units as the gate capacitances (i.e., multiples of the capacitance of a unit inverter). Assume the inputs arrive at time 0. Write an expression for the arrival time of the output as a function of the gate drives. Form the optimization problem to minimize delay.



Question 5

Sketch a 2-input NOR gate with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance.

Question 6

Find the rising and falling propagation delays of an unloaded AND-OR-INVERT gate using the Elmore delay model. Estimate the diffusion capacitance based on a stick diagram of the layout.

Question 7

Find the worst-case Elmore parasitic delay of an n-input NOR gate.

Question 8

Let a 4x inverter have transistors four times as wide as those of a unit inverter. If a unit inverter has three units of input capacitance and parasitic delay of ρ_{inv} , what is the input capacitance of a 4x inverter? What is the logical effort? What is the parasitic delay?

Question 9

Consider a process in which pMOS transistors have three times the effective resistance as nMOS transistors. Calculate the logical efforts of a 2-input NAND gate and a 2-input NOR gate if they are designed with equal rising and falling delays.